

Syncing with mobile devices

In a way, the growth of wearable technology products can be attributed to the popularity of smartphones and tablets. While wearable products are capable of doing a certain amount of standalone processing, most of the current generation products work together with a mobile device—which gives them access to the whole Cloud! They simply partner with a connected mobile device via Wi-Fi, Bluetooth or other wireless communications technology, and transfer a large part of the processing and value-add work to the mobile. In fact, this is what enables the creation of light and safe wearable technology products that can still perform meaningful tasks.

Wireless technologies like 3G/4G, GPS and Bluetooth are popular in wearable devices like watches and augmented-reality glasses. While these are relatively mature technologies, the actual challenge in using these in wearable devices is three-fold: Enabling these on the device using very little power; coping with the general telecom issues that plague all the mobile devices; and selecting the right platform and software in order to handle such data transactions securely.

"The features of today's wearable products are limited to basic connectivity like Wi-Fi and Bluetooth. Battery life is one of the most important considerations for wearable devices—much like it is for mobile devices. Even technologies and hardware that support communication protocols like Bluetooth and 3G radio are battery-intensive," says Vishwanathan.

Vishwanathan adds "Communication of information collected from the wearable devices to another device or to servers is also tricky. Connectivity issues like keep-alive connections, connection drops, and switching between cell towers and others systems are some of the most challenging issues to manage even in traditional mobile applications, and these challenges carry forward to wearable devices as well."

Another important consideration is to fit the relevant information into

the tiny screens of wearable devices. Traditional, non-software devices are able to do it elegantly, but software as a paradigm is data-intensive, and simplifying that information and interface to fit into smaller screens is something that software vendors have to ensure.

Recently, Persistent Systems released a white-paper on Consumerware—a form of consumerisation of software—that talks about ten characteristics of Consumerware which traditional software might lack. At a high level, Consumerware applications are endpoint agnostic, context-driven as opposed to event-driven, content-centric rather than program-centric, transformational instead of transactional, data-driven, collaborative, and will bridge the final frontier in human-machine interaction. So mobile devices including wearable ones are actually leading to a disruptive change in the software world.

Ganesan adds, "In fact, with the help of communication technologies, one can use their wearable devices to connect to other devices and hence enable the Internet of Things."

Testing to ensure compliance

"Electronic devices radiate RF power, which could be dangerous for the human body. Imagine a piece of chicken in a microwave oven. It gets cooked. The same is bound to happen when excess RF power is kept close to a human body. There are compliance standards like CE to regulate this, and compliance testing for electromagnetic interference/compatibility (EMI/EMC) is one of the major considerations while developing wearable technology," says Satish Mohanram, technical marketing manager, National Instruments-India.

Temperature profile is another key consideration. The heating effect of the system has to be studied in order to understand the hotspots and one has to make sure that heat is properly dissipated. Then there are shock and vibration tests that test the ruggedness of the device—which is very important as the wearer might unknowingly drop or hit the object against tough surfaces.

How reliable are these devices?

Endpoint security and privacy are very important. Many of the wearable devices are regulated as part of industries like healthcare.

"In healthcare, for example, the devices have to adhere to HIPAA regulations when storing and communicating patient's information. These regulations apply to mobile devices. Companies building solutions around wearable devices can look at how traditional mobile applications and solutions overcome them, and apply them into their setting," says Vishwanathan.

Data encryption and establishing secure communication channels will ensure that sensitive data in storage as well as transit is adequately secure. Privacy is a bit more tricky and involves how and where the data is stored and who (including which other applications) has access to it once it reaches a server, what data from a wearable device is being 'recorded' or used, and whether the user has given explicit consent to that usage.

"In general, we are witnessing consumerisation of software at multiple levels on mobile devices and wearable devices. So privacy concerns will be discussed a lot in future, similar to what we have seen on consumer Web applications like Facebook and some Google products," adds Vishwanathan.

What's being worn now?

Let us now look at some interesting examples of wearable devices that are in the making or available in the market today.

Lifestyle monitors. Physical activity monitors like Fitbit, Larklife and Nike's Fuelband are very popular today. All are wristbands with built-in triaxial accelerometers. While Fuelband monitors only physical activity, Fitbit and Larklife also track diet and sleep. Such devices are powered by Lithium-polymer batteries, and connect to other devices using means like Bluetooth and USB 2.0.

The devices are coupled with an application running on a mobile phone,